PROPOSITION INJECTIVE IN PIONE.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter pione.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:pione.tex:proposition-injective.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0BN7: sources/pione.tex, source lines 858–858. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BTV: sources/pione.tex, source lines 3428–3428. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0G1H: sources/pione.tex, source lines 7584–7584. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let p be a prime number. Let $i : Z \to X$ be a closed immersion of connected affine schemes over $\mathbf{F}_p$. For any geometric point $\overline{z}$ of Z the map*

$$\pi_1(Z, \overline{z}) \to \pi_1(X, \overline{z})$$

is injective.

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Proof. Let $Y \to Z$ be a finite étale morphism. It suffices to construct a finite étale morphism $f : U \to X$ such that Y is isomorphic to an open and closed subscheme of $f^{-1}(Z)$, see Lemma 0BN7. Write $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(R)$ so the closed immersion $Y \to X$ is given by a surjection $R \to A$. We may write $A = \mathrm{colim}\, A_i$ as the filtered colimit of its $\mathbf{F}_p$-subalgebras of finite type. By Lemma 0BTV we can find an i and a finite étale morphism $Y_i \to Z_i = \mathrm{Spec}(A_i)$ such that $Y = Z \times_{Z_i} Y_i$. Choose a surjection $\mathbf{F}_p[x_1, \ldots, x_n] \to A_i$. This determines a closed immersion

$$Z_i = \mathrm{Spec}(A_i) \longrightarrow X_i = \mathbf{A}^n_{\mathbf{F}_p} = \mathrm{Spec}(\mathbf{F}_p[x_1, \ldots, x_n])$$

By the universal property of polynomial algebras and since $R \to A$ is surjective, we can find a commutative diagram

$$\begin{array}{ccc} \mathbf{F}_p[x_1, \ldots, x_n] & \longrightarrow & A_i \\ \downarrow & & \downarrow \\ R & \longrightarrow & A \end{array}$$

of $\mathbf{F}_p$-algebras. Thus we have a commutative diagram

$$\begin{array}{ccccc} Y_i & \longrightarrow & Z_i & \longrightarrow & X_i \\ \uparrow & & \uparrow & & \uparrow \\ Y & \longrightarrow & Z & \longrightarrow & X \end{array}$$

whose right square is cartesian. Clearly, if we can find $f_i : U_i \to X_i$ finite étale such that Y_i is isomorphic to an open and closed subscheme of $f_i^{-1}(Z_i)$, then the base change $f : U \to X$ of f_i by $X \to X_i$ is a solution to our problem. Thus we conclude by applying Lemma 0G1H to $Y_i \to Z_i \to X_i = \mathbf{A}^n_{\mathbf{F}_p}$. □